WEEK 6 — SPINOFF ARBITRAGE PROJECT · 1 OF 2

How the Classification Tree Works

44%

nested LOO-CV accuracy, leakage-free · vs. 38% majority-class baseline

The tree predicts which tercile of the 252d parent CAR distribution an event lands in — `underperform` / `middle` / `outperform` — from four features: `prob_included`, `forced_flow_adv`, `log_child_mktcap`, `beta`. Selection and scoring both use nested LOO-CV so no decision ever sees the label it's predicting. n = 16 events.

HOW IT'S FIT: NESTED LOO-CV + A 3-LEAF FLOOR

LEAKAGE-FREE

```
for each held-out event (outer LOO):
  grow full tree on the other 15 → candidate ccp_alphas
  keep only alphas whose tree has ≥ 3 leaves
    (a 2-leaf tree can't output a 3rd class — floor makes
     "underperform" structurally reachable)
  inner LOO over those 15 picks the best-scoring alpha
  refit on all 15 → predict the held-out event
```

— Every outer fold picked `alpha=0` — at n=16, the ≥3-leaf floor already excludes every more-pruned candidate, so the search confirms the full, unconstrained tree is the only option that qualifies.

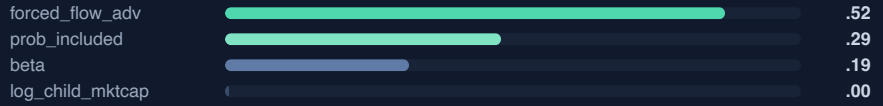
MAJORITY BASELINE	NESTED LOO ACCURACY
38%	44%

— "Underperform" recall: **0%** in last week's 2-leaf tree → **40%** now — the floor is what makes flagging the downside case possible at all.

THE LEARNED TREE (FIT ON ALL 16 EVENTS)

DEPTH 3 · 6 LEAVES

```
├── forced_flow_adv ≤ 7.43
│   ├── prob_included ≤ 0.62
│   │   ├── beta ≤ 0.98 → middle
│   │   └── beta > 0.98 → underperform
│   └── prob_included > 0.62 → middle
└── forced_flow_adv > 7.43
    ├── prob_included ≤ 0.36 → underperform
    └── prob_included > 0.36
        ├── forced_flow_adv ≤ 18.35 → outperform
        └── forced_flow_adv > 18.35 → underperform
```



	pred: under	pred: mid	pred: over
actual: under	2	1	2
actual: mid	2	3	1
actual: over	2	1	2

Nested LOO-CV predictions, all 16 events. Diagonal = correct. Forced-selling flow (`forced_flow_adv`) drives most splits — more than `prob_included` alone.

n=16 Read as hypothesis, not strategy. **44% vs. 38%** is a real, leakage-free improvement — but it comes from one tree shape confirmed across all 16 outer folds, not a genuine complexity search. Next: how this tree's predictions turn into a position-sizing weight (Slide 2 of 2).